

Wind Estimate Uncertainty Quantification and Sensitivity Analysis Using Generalized Polynomial Chaos

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<https://doi.org/10.2514/1.G008513>

Indirect wind estimation onboard unmanned aerial systems (UASs) can be accomplished using existing onboard sensors along with a dynamic model of the UAS augmented with additional wind-related states. Although many indirect estimation approaches achieve desirable accuracy under the assumption of known system parameters, the effect of parametric uncertainty on wind estimate precision is important and has not been thoroughly investigated. This paper describes the theory and process for using generalized polynomial chaos (gPC) to recast the dynamics of a system with nondeterministic parameters as a deterministic system. The concepts are applied to the problem of wind estimation and to characterizing the precision of wind estimates over time due to known parametric uncertainties. In the multivariate case, gPC exhibits a computational advantage over Monte Carlo sampling-based methods and produces a deterministic system upon which control and estimation methods can be applied.

I. Introduction

WIND estimation from aircraft motion is useful for many reasons, including weather forecasting, air traffic management, and feedback control [1]. The ability to estimate wind indirectly (i.e., without specialized instrumentation) is of particular interest due to the increasing ubiquity of small unmanned aerial systems (UASs) [2] and the unobtrusive/low-cost nature of using operational sensors [e.g., air vehicle Global Positioning System (GPS), accelerometers, and gyroscopes]. Generally, these estimation methods can be grouped into two major categories: model-based and model-free. Model-based estimation requires accurate aerodynamic and propulsion models of the aircraft to generate state and wind estimates [3,4]. Model-based methods are popular because they incorporate knowledge of the vehicle dynamics. They do not require direct measurement of the wind, such as with an air data sensor [5], and are thus readily applied to UASs. However, the current literature often does not address uncertainty in wind estimates due to uncertainties in the aircraft model.

Uncertainty refers to how imperfect knowledge is manifested in a system or process model [6]. Uncertainty is generally categorized as either aleatory uncertainty, which is inherent (irreducible) randomness in a system or process, or epistemic uncertainty, which is incomplete or inexact (reducible) encoding of the physical system or process into the model [7].

Uncertainty quantification (UQ) is the science of characterizing the uncertainty of model outputs due to uncertain inputs and assumptions ([8], Ch. 1). UQ is useful in determining the confidence one should place in the results of a simulation or experiment. Traditionally, sampling-based methods, such as Monte Carlo (MC) analysis, have been the primary tool for UQ [9,10]. In MC analysis, numerous deterministic simulations are used to generate posterior statistical information based on known or assumed a priori probability distributions for the uncertain elements of a system. MC

for UQ has been applied to many diverse problems, such as flutter analysis [11], entry vehicle trajectory prediction [12], and chemical reactions [13], to name a few. Although MC is a reliable method for UQ and is often a baseline for other methods, it is computationally expensive, sometimes prohibitively so [11]. To avoid the computational burden of MC analysis, approaches such as design of experiments/response surface methodology (DOE/RSM) are available, in which an approximated input/output model is tested with a minimal set of inputs [11]. However, this method is highly dependent on the proper choice of response surface equation and can yield inaccurate or false information if the equation is not chosen carefully. Neither MC nor DOE/RSM are tractable for real-time applications. As an alternative, we present a method for characterizing the aleatory uncertainty in wind estimates due to imperfect knowledge of system parameters, which are treated as stochastic, using generalized polynomial chaos (gPC) [14] to overcome these limitations.

A field closely related to UQ is sensitivity analysis (SA). SA seeks to determine how the model output uncertainty can be apportioned to individual input (parametric) uncertainties or combinations thereof [15]. Ideally, UQ and SA are performed concurrently to evaluate model confidence and identify areas for model improvement. As with UQ, MC is the most commonly used method for conducting SA due to the ease of implementation [16–18]. Again, this comes with a significant computational cost to conduct the simulations necessary to achieve reliable results. We present the use of gPC for wind estimate SA, along with a process for reducing input complexity, which is computationally superior to MC.

In Sec. II, we present the aircraft motion model for which we develop the state estimator. In Sec. III, we develop the methods for analyzing uncertainty in wind estimates. Section VI presents an example using flight-test data and multivariate uncertainty. In Sec. VII, we present the results from multivariate analysis. Finally, in Sec. VIII, we present conclusions.

II. Aircraft Motion in Wind

In developing the aircraft motion model for model-based estimation, we begin by defining a wind field. We then present a nonlinear flight dynamic model, with a common simplification (Assumption 1) for the effect of wind on the aircraft. Finally, we linearize the flight dynamic model about an arbitrary steady motion for use in the wind estimator design and analysis. Although these methods could be applied to any aircraft, this paper will focus on fixed-wing aircraft.

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A. Wind Field

Consider an aircraft in a wind velocity field $\mathbf{W}$ that varies in space and time. At an instant of time, the wind velocity that would exist at the aircraft center of gravity (c.g.) if the aircraft were absent (the *apparent wind*) is

$$\mathbf{w}(t) = \mathbf{W}(\mathbf{X}(t), t) \quad (1)$$

where $\mathbf{X} = [X_N \ X_E \ X_D]^\top$ is the north–east–down position of the aircraft c.g. in a reference frame $\mathcal{F}_I$. Although $\mathcal{F}_I$ is fixed to the Earth's surface, we consider it to be an inertial reference frame, as is commonly done over short time intervals. We define it using the orthonormal triad $\{\mathbf{i}_1, \mathbf{i}_2, \mathbf{i}_3\}$. The total time derivative of the apparent wind is

$$\frac{d\mathbf{w}}{dt} = \frac{\partial \mathbf{W}(\mathbf{X}, t)}{\partial t} + \frac{\partial \mathbf{W}(\mathbf{X}, t)}{\partial \mathbf{X}} \frac{d\mathbf{X}}{dt} \quad (2)$$

Adopting the standard overdot notation for the total time derivative and suppressing the arguments gives

$$\dot{\mathbf{w}} = \frac{\partial \mathbf{W}}{\partial t} + \frac{\partial \mathbf{W}}{\partial \mathbf{X}} \dot{\mathbf{X}} \quad (3)$$

B. Aircraft Nonlinear Equations of Motion in Wind

Having introduced the dynamics of the wind field in Sec. II.A, we now briefly present the aircraft equations of motion relative to the wind. The reader is directed to [19,20] for a full treatment of the derivation. Begin by defining the orthonormal aircraft body reference frame $\mathcal{F}_B$ defined by unit vectors $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ originating at the c.g., where $\mathbf{b}_1$ points forward, $\mathbf{b}_2$ points out the right wing, and $\mathbf{b}_3$ completes the right-handed reference frame. The transformation mapping free vectors from $\mathcal{F}_B$ to $\mathcal{F}_I$ is given by the rotation matrix

$$\mathbf{R}_{IB}(\boldsymbol{\Theta}) = e^{[e_3 \times] \psi} e^{[e_2 \times] \theta} e^{[e_1 \times] \phi} \quad (4)$$

where $\boldsymbol{\Theta} = [\phi \ \theta \ \psi]^\top$ is the vector comprising the aircraft roll, pitch, and yaw Euler angles, respectively. Here, $\mathbf{e}_1 = [1 \ 0 \ 0]^\top$, etc., and $[(\cdot) \times]$ is the skew-symmetric cross-product equivalent matrix satisfying $[\mathbf{a} \times] \mathbf{b} = \mathbf{a} \times \mathbf{b}$ for three-vectors $\mathbf{a}$ and $\mathbf{b}$. $\mathbf{R}_{BI} := \mathbf{R}_{IB}^{-1}$ because $\mathbf{R}_{IB}$ is orthonormal. The aircraft kinematics are given by

$$\dot{\mathbf{X}} = \mathbf{R}_{IB} \mathbf{v} \quad (5)$$

$$\dot{\mathbf{R}}_{IB} = \mathbf{R}_{IB} [\boldsymbol{\omega} \times] \quad (6)$$

where $\mathbf{v} = [u \ v \ w]^\top$ is the aircraft velocity with respect to the inertial frame, and $\boldsymbol{\omega} = [p \ q \ r]^\top$ is the aircraft angular velocity with respect to the inertial frame, where both $\mathbf{v}$ and $\boldsymbol{\omega}$ are expressed in the body frame. Although the matrix differential equation (6) holds for any parameterization of $\mathbf{R}_{IB}$, adopting Euler angles gives the more compact system

$$\dot{\boldsymbol{\Theta}} = \mathbf{L}_{IB}(\boldsymbol{\Theta}) \boldsymbol{\omega} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi \sec \theta & \cos \phi \sec \theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad (7)$$

The aerodynamic forces $\mathbf{F}$ and moments $\mathbf{M}$ depend on the velocity of the aircraft relative to the wind. Let

$$\mathbf{v}_r = \mathbf{v} - \mathbf{R}_{BI}(\boldsymbol{\Theta}) \mathbf{w} \quad (8)$$

where $\mathbf{v}_r = [u_r \ v_r \ w_r]^\top$ is the velocity of the body's center of gravity with respect to the wind, given in the body frame. Equation (8) is known as the wind triangle. To simplify the derivation of the aircraft equations of motion, the following assumption is adopted:

Assumption 1: The flow is steady and uniform over the scale of the aircraft; that is, $(\partial \mathbf{W} / \partial t)$ and $(\partial \mathbf{W} / \partial \mathbf{X})$ are ignored in the aircraft dynamics.

Assumption 1 may seem incompatible with the motivating challenge of estimating variations in the wind velocity. However, the assumption does not impede the model-based filter from adjusting wind velocity estimates based on measurements. This assumption, which is consistent with [1,21], could be relaxed, but it simplifies the results presented here. One consequence of Assumption 1 is that the air-relative angular velocity is equal to the vehicle angular velocity $\boldsymbol{\omega}$. We therefore assume that the aerodynamic forces and moments have the general dependence:

$$\mathbf{F} = \mathbf{F}(\mathbf{v}_r, \boldsymbol{\omega}, \mathbf{u}), \quad \mathbf{M} = \mathbf{M}(\mathbf{v}_r, \boldsymbol{\omega}, \mathbf{u}) \quad (9)$$

where $\mathbf{u} = [\delta_T \ \delta_a \ \delta_e \ \delta_r]^\top$ are the throttle, aileron, elevator, and rudder control inputs, respectively.

In this work, we consider a particular aircraft model as a representative example. The aerodynamic model is the one described in [22]. The model has nondimensional coefficients of the form

$$C_X = C_{X_o} + C_{X_\alpha} \alpha + C_{X_{\alpha^2}} \alpha^2 + C_{X_{\delta_e}} \delta_e + C_T \delta_T \quad (10a)$$

$$C_Z = C_{Z_o} + C_{Z_\alpha} \alpha + C_{Z_q} \hat{q} \quad (10b)$$

$$C_m = C_{m_o} + C_{m_\alpha} \alpha + C_{m_q} \hat{q} + C_{m_{\delta_e}} \delta_e \quad (10c)$$

$$C_Y = C_{Y_\beta} \beta + C_{Y_p} \hat{p} + C_{Y_r} \hat{r} + C_{Y_{\delta_a}} \delta_a + C_{Y_{\delta_r}} \delta_r \quad (10d)$$

$$C_l = C_{l_\beta} \beta + C_{l_p} \hat{p} + C_{l_{\delta_a}} \delta_a \quad (10e)$$

$$C_n = C_{n_\beta} \beta + C_{n_r} \hat{r} + C_{n_{\delta_a}} \delta_a + C_{n_{\delta_r}} \delta_r \quad (10f)$$

where the propulsion coefficient C_T is modeled using information from [23]. Here, an overhat ($\hat{\cdot}$) indicates a nondimensional angular rate. (The same notation will later be used to denote an estimate of a state variable.) The angle of attack α and angle of sideslip β are defined as follows:

$$\alpha = \tan^{-1} \left(\frac{w_r}{u_r} \right) \quad \text{and} \quad \beta = \sin^{-1} \left(\frac{v_r}{V_r} \right) \quad (11)$$

where $V_r = \|\mathbf{v}_r\|$. Note that Eqs. (9) and (10) omit the common quasi-steady approximations of unsteady effects (e.g., $\dot{\alpha}$ and $\dot{\beta}$). Thus, the well established nonlinear equations of motion, covered at length in [19,24], adapted for the assumed wind field, are

$$\dot{\mathbf{X}} = \mathbf{R}_{IB}(\boldsymbol{\Theta}) \mathbf{v}_r + \mathbf{w} \quad (12a)$$

$$\dot{\boldsymbol{\Theta}} = \mathbf{L}_{IB}(\boldsymbol{\Theta}) \boldsymbol{\omega} \quad (12b)$$

$$\dot{\mathbf{v}}_r = \mathbf{v}_r \times \boldsymbol{\omega} + g \mathbf{R}_{BI}(\boldsymbol{\Theta}) \mathbf{e}_3 + \frac{1}{m} \mathbf{F}(\mathbf{v}_r, \boldsymbol{\omega}, \mathbf{u}) \quad (12c)$$

$$\dot{\boldsymbol{\omega}} = \mathbf{I}^{-1} (\mathbf{I} \boldsymbol{\omega} \times \boldsymbol{\omega} + \mathbf{M}(\mathbf{v}_r, \boldsymbol{\omega}, \mathbf{u})) \quad (12d)$$

where $\mathbf{I}$ is the matrix of aircraft moments and products of inertia, as computed in $\mathcal{F}_B$. The aircraft state and measurement equations are

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, \mathbf{w}) \quad (13a)$$

$$\mathbf{y} = \mathbf{h}(\mathbf{x}, \mathbf{u}, \mathbf{v}) \quad (13b)$$

where the state, input, and wind vectors are

$$\mathbf{x} = \begin{bmatrix} \mathbf{X}^\top & \boldsymbol{\Theta}^\top & \mathbf{v}_r^\top & \boldsymbol{\omega}^\top \end{bmatrix}^\top, \quad \mathbf{u} = \begin{bmatrix} \delta_T & \delta_a & \delta_e & \delta_r \end{bmatrix}^\top, \\ \mathbf{w} = \begin{bmatrix} w_N & w_E & w_D \end{bmatrix}^\top \quad (14)$$

and where ν is an unknown finite-energy measurement disturbance (i.e., noise). We now present the linearized equations of motion for the aircraft in a wind field.

C. Linearized Small-Perturbation Equations for Fixed-Wing Aircraft Motion in Wind

For the purpose of wind estimation, the aircraft state of motion is assumed to remain in the neighborhood of a nominal equilibrium flight condition over a desired time or distance. Recall that the most general “aerodynamic steady state” ([19], sec. 5.9) is a vertical helix in calm air. This nominal flight condition is defined with regard to the mean wind field. Let the subscript $\mathbf{eq} = (\mathbf{x}_e, \mathbf{u}_e, \mathbf{w}_e)$ denote the state and control variables and the mean wind velocity corresponding to a given equilibrium flight condition (e.g., straight and level cruise in still air). Having defined a nominal equilibrium state of motion and assuming that the flight dynamic model is exact, any perturbations from the equilibrium flight condition that are not attributed to the initial state or control inputs can be ascribed to the wind disturbance.

To prepare for the formulation of a linear filter, it is necessary to develop a linear time-invariant (LTI) model from the equations of motion given in Sec. II.B. Define

$$\mathbf{A} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\mathbf{eq}} \quad \mathbf{B} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right|_{\mathbf{eq}} \quad \boldsymbol{\Pi} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{w}} \right|_{\mathbf{eq}} \quad (15)$$

Linearizing the time-invariant system [Eq. (12)] about the equilibrium flight condition gives the LTI system

$$\Delta \dot{\mathbf{x}}(t) = \mathbf{A} \Delta \mathbf{x}(t) + \mathbf{B} \Delta \mathbf{u}(t) + \boldsymbol{\Pi} \Delta \mathbf{w}(t), \quad \Delta \mathbf{x}(0) = \Delta \mathbf{x}_0 \quad (16a)$$

$$\Delta \mathbf{y}(t) = \mathbf{C} \Delta \mathbf{x}(t) + \mathbf{D} \Delta \mathbf{u}(t) + \boldsymbol{\Lambda} \nu(t) \quad (16b)$$

respectively. Additionally, $\Delta \mathbf{y}(t) \in \mathbb{R}^{n_y}$ is the measurement perturbation, $\mathbf{C}$ and $\mathbf{D}$ are the output and feedthrough matrices, respectively, and $\boldsymbol{\Lambda}$ is the measurement noise input matrix. Note that, although the position and heading variables in $\mathbf{x}_e$ depend on time, they are ignorable coordinates in the dynamic equations, and the resulting small-perturbation model is LTI. For brevity, we drop the Δ notation.

III. Wind Estimation and Uncertainty

This section describes the process of using gPC to characterize the aleatory uncertainty in wind estimates due to the stochastic nature of the linear model matrices. As described in Sec. II, the typical process for developing the linear aircraft model needed for wind estimation involves first postulating a nonlinear model structure, then identifying the model parameters from experimentation, and finally linearizing about the desired condition. During the system identification process, parametric uncertainty can largely be viewed as epistemic. That is, by gathering additional data or considering different model structures, we could reduce model uncertainty with additional time and effort. Commonly used system identification processes [25] (Chap. 6) aim at determining a model structure and model parameters for which the model prediction error is a minimum-variance uncorrelated random process. The error is typically attributed to random disturbances and measurement noise in the output data used for model identification, but a direct consequence is a nondeterministic model in which the model parameters are described by probability distributions (e.g., mean and variance). For this reason, we consider the parameter uncertainty to be well described as aleatory.

A. Augmented System and Filtering

To generate wind estimates, we augment the aircraft state vector with the wind states, following the approach in [21], defining $\mathbf{x}_A = [\mathbf{x}^\top \mathbf{w}^\top]^\top$ so that

$$\dot{\mathbf{x}}_A(t) = \begin{bmatrix} \mathbf{A} & \boldsymbol{\Pi} \\ \mathbf{0} & * \end{bmatrix} \mathbf{x}_A(t) + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{u}(t) = \mathbf{A}_A \mathbf{x}_A(t) + \mathbf{B}_A \mathbf{u}(t) \quad (17a)$$

$$\mathbf{y}_A(t) = [\mathbf{C} \quad \mathbf{0}] \mathbf{x}_A(t) + \mathbf{D} \mathbf{u}(t) + \boldsymbol{\Lambda} \nu(t) = \mathbf{C}_A \mathbf{x}_A(t) + \mathbf{D} \mathbf{u}(t) + \boldsymbol{\Lambda} \nu(t) \quad (17b)$$

where $*$ is based on the user’s model of the wind dynamics. (For example, in [26,27], a low-pass filter was used.) A Luenberger filter for the augmented system of Eq. (17) is of the form

$$\dot{\hat{\mathbf{x}}}_A(t) = \mathbf{A}_A \hat{\mathbf{x}}_A(t) + \mathbf{B}_A \mathbf{u}(t) + \mathbf{K}(\mathbf{y}_A(t) - \mathbf{C}_A \hat{\mathbf{x}}_A(t)), \quad \hat{\mathbf{x}}_A(0) = \hat{\mathbf{x}}_{A_0} \quad (18a)$$

$$\hat{\mathbf{z}}(t) = \mathbf{L} \hat{\mathbf{x}}_A(t) \quad (18b)$$

where the overhat notation ($\hat{\cdot}$) now denotes an estimate, $\hat{\mathbf{z}}$ are the estimates of interest (e.g., the components of the wind velocity), and $\mathbf{K}$ is a generic filter gain matrix determined using a method such as KALMAN–BUCY filtering or H_∞ filtering [26,27].

B. Mapping from Nonlinear Uncertainty to Linear Uncertainty

We first note that the nonlinear equations $\mathbf{f}$ can be separated into kinematic and dynamic equations, $\mathbf{f} = [\mathbf{f}_k^\top \mathbf{f}_d^\top]^\top$, respectively. In this study, we seek to address the uncertainty in the dynamic equations $\mathbf{f}_d$ that arises from the aerodynamic parameters ($\boldsymbol{\theta} \in \mathbb{R}^m$) given in Table A1 in Appendix A. The aerodynamic model [Eq. (10)] is linear in its parameters. This is a common choice of model structure [25] (Chap. 4). Thus, the dynamic equations can be further separated into parts independent of and dependent on $\boldsymbol{\theta}$:

$$\mathbf{f}_d(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \bar{\mathbf{f}}_d(\mathbf{x}, \mathbf{u}) + \sum_{i=1}^m \bar{\mathbf{g}}_i(\mathbf{x}, \mathbf{u}) \vartheta_i \quad (19)$$

We can similarly separate $\mathbf{A}$, $\mathbf{B}$, and $\boldsymbol{\Pi}$ into kinematic and dynamic components:

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_k \\ \mathbf{A}_d \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \mathbf{B}_k \\ \mathbf{B}_d \end{bmatrix}, \quad \boldsymbol{\Pi} = \begin{bmatrix} \boldsymbol{\Pi}_k \\ \boldsymbol{\Pi}_d \end{bmatrix} \quad (20)$$

Then, from Eqs. (15) and (19), the elements of $\mathbf{A}_d$ are

$$A_{d_{p,q}} = \left. \frac{\partial \bar{f}_{d_p}}{\partial x_q} \right|_{\mathbf{eq}} + \sum_{i=1}^m \left. \frac{\partial \bar{g}_{i,p}}{\partial x_q} \right|_{\mathbf{eq}} \vartheta_i \quad (21)$$

The elements of $\mathbf{B}_d$ and $\boldsymbol{\Pi}_d$ are found similarly. Equation (21) gives an expression for how uncertainty in the parameters of the nonlinear model maps to uncertainty in the linear system. To simplify the presentation of gPC, we make the following common assumption [28–30]:

Assumption 2: The nonlinear model parameters are statistically independent [31].

Under Assumption 2, we can use Eq. (21), along with knowledge of the characteristics of the uncertainty of the nonlinear model parameters, to determine the linear model uncertainties. Referring again to Table A1, the result of the system identification process is that the nonlinear model parameters are normally distributed, $\vartheta_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, where μ is the mean value, and σ is the standard deviation. Thus, we can apply the normal sum theorem [31] to get

$$\mu(A_{d_{p,q}}) = \left. \frac{\partial \bar{f}_{d_p}}{\partial x_q} \right|_{\text{eq}} + \sum_{i=1}^m \left. \frac{\partial \bar{g}_{i,p}}{\partial x_q} \right|_{\text{eq}} \mu(\vartheta_i) \quad (22)$$

$$\sigma^2(A_{d_{p,q}}) = \sum_{i=1}^m \left(\left. \frac{\partial \bar{g}_{i,p}}{\partial x_q} \right|_{\text{eq}} \sigma(\vartheta_i) \right)^2 \quad (23)$$

The resulting equations are provided in Appendix B. We now have

$$A(\chi) = \begin{bmatrix} A_k \\ A_d(\xi) \end{bmatrix} \quad (24)$$

where ξ are canonical random variables known as germs, described further in Sec. III.C.1, and n_ξ is the number of germs. Finally, $\chi \in \mathbb{R}^{n_\xi}$ is the vector of random parameters for the linear system. Following a similar process, we can find $B(\chi)$ and $\Pi(\chi)$.

C. Uncertain Filter Equations

Consider the uncertain form of the filter equations (18):

$$\dot{\hat{x}}_A(t, \chi) = A_A(\chi) \hat{x}_A(t, \chi) + B_A(\chi) u(t) + K(y_A(t) - C_A \hat{x}_A(t, \chi)) \quad (25a)$$

$$\hat{z}(t, \chi) = L \hat{x}_A(t, \chi) \quad (25b)$$

Because A_A and B_A are functions of χ , it follows that $\hat{x}_A$ and $\hat{z}$ are stochastic. The matrices C_A , K , and L are deterministic. Specifically, C_A and L are determined by available measurements and desired estimates, respectively, and K is designed using some nominal deterministic system.

For these stochastic LTI equations, we use gPC to perform generalized polynomial chaos expansion (gPCE), which results in a deterministic system of equations. We then use the resulting gPC system to characterize uncertainty in the wind estimates.

1. Physical Random Variables and gPC Germs

Before proceeding, it is worthwhile to think of the random vector χ as a vector of physical random variables. The characteristics of these physical random variables (distribution, mean, variance, etc.) are the results of observations or assumptions regarding tangible aspects of the aircraft. For example, if we were considering uncertainty in the aircraft mass, then the mean would certainly be some nonzero positive value. As mentioned at the beginning of this chapter, the current uncertainty we are addressing is tied to the system identification methods used. However, the system identification coefficients and regressors [25] (Chap. 5) also have physical meaning.

Alternatively, germs ξ are canonical random variables that have a one-to-one correspondence with certain families of orthogonal polynomials [32]. The polynomials are derived from the Askey scheme, and a germ-polynomial pair is known as WIENER-ASKEY polynomial chaos [32]. Note that gPC is performed around germs and not around physical random variables directly. For example, a uniform random variable $\chi \sim \mathcal{U}(5, 10)$ requires support on the interval [5, 10]; however, the support for the uniform germ is $[-1, 1]$. Thus, it is necessary to recast the physical random vector χ in terms of ξ . This recasting process may be accomplished via a gPCE or through some other transformation [8, 33, 34].

To illustrate, let us assume that there is a normally distributed parameter $c(\chi_1)$ with mean $\mu(c)$ and standard deviation $\sigma(c)$. First, we need to recast $c(\chi_1)$ in terms of a germ so that we can apply gPC. Since $c(\chi_1)$ is normally distributed, the GAUSSIAN germ is appropriate. Following [33] (sec. 5.1.2)

$$c(\chi_1) = \mu(c) + \sigma(c)\xi_1 \quad (26)$$

where $\xi_1 \sim \mathcal{N}(0, 1)$. A thorough demonstration of this process is given in [35].

As we proceed further with the discussion of gPC, the uncertain filter equations [Eq. (25)] are assumed to be recast using germs:

$$\dot{\hat{x}}_A(t, \xi) = A_A(\xi) \hat{x}_A(t, \xi) + B_A(\xi) u(t) + K(y_A(t) - C_A \hat{x}_A(t, \xi)) \quad (27a)$$

$$\hat{z}(t, \xi) = L \hat{x}_A(t, \xi) \quad (27b)$$

D. Generalized Polynomial Chaos

Let us define a tuple $j = (j_1, \dots, j_{n_\xi}) \in \mathbb{N}_0^{n_\xi}$, where $\mathbb{N}_0$ are the natural numbers including 0, with $|j| = j_1 + \dots + j_{n_\xi}$ denoting the total degree of j [36]. We can construct multivariate gPC basis functions as products of univariate gPC polynomials according to the WIENER-ASKEY scheme [37]; that is

$$\phi_j(\xi) = \phi_{j_1}(\xi_1) \dots \phi_{j_{n_\xi}}(\xi_{n_\xi}), \quad 0 \leq |j| \leq \infty \quad (28)$$

From [32], a second-order random process (i.e., finite variance) can be represented as a gPCE. Thus, for a function $\psi(\xi) : \mathbb{R}^{n_\xi} \rightarrow \mathbb{R}$,

$$\psi(\xi) = \sum_{|j|=0}^{\infty} \psi_j \phi_j(\xi) \quad (29)$$

where $\{\psi_j\}$ are the set of expansion coefficients. This expansion converges in the $\mathcal{L}_2$ sense [32, 38]. Fundamental to the concept of gPCE is the orthogonality of basis functions with respect to the distribution $\varphi(\xi)$ of ξ , where $\varphi(\xi) = \prod_{k=1}^{n_\xi} \varphi_k(\xi_k)$. That is

$$\langle \phi_j(\xi), \phi_k(\xi) \rangle := \int_{\Omega} \phi_j(\xi) \phi_k(\xi) \varphi(\xi) d\xi \quad (30a)$$

$$= \mathbb{E}_{\xi} \{ \phi_j(\xi) \phi_k(\xi) \} \quad (30b)$$

$$= \begin{cases} \langle \phi_j(\xi), \phi_j(\xi) \rangle, & \text{if } j = k \\ 0, & \text{if } j \neq k \end{cases} \quad (30c)$$

where Ω is the support of $\varphi(\xi)$ given by $\Omega = \Omega_1 \times \dots \times \Omega_{n_\xi}$ [37]. In practice, it is not feasible to perform an exact gPCE because this requires an infinite number of basis functions. Instead, the gPCE is truncated to some suitably high order such that the total degree $|j| \leq p$:

$$\psi(\xi) \approx \sum_{|j|=0}^p \psi_j \phi_j(\xi) =: \tilde{\psi}(\xi) \quad (31)$$

where p is the polynomial order, and the number of expansion factors is

$$N_p = \binom{n_\xi + p}{p} := \frac{(n_\xi + p)!}{p!(n_\xi + p - p)!} = \frac{(n_\xi + p)!}{n_\xi! p!} \quad (32)$$

For convenient manipulation of the multi-index j , the graded lexicographic order (GLO) is often used [36], where $k > j$ if $|k| > |j|$, or if $|k| = |j|$ and the first nonzero entry in $k - j$ is positive. Table 1 presents the resulting ordering of a 3-tuple, as well as the associated single-index j' . The orthogonality property expressed in Eq. (30) ensures that $\tilde{\psi}(\xi)$ is the minimum error (ϵ) representation of $\psi(\xi)$ with respect to the Euclidean norm on the space spanned by the truncated basis functions [39]. This is known as a GALERKIN projection [39]. Figure 1 shows this concept for a two-term gPCE in a single uncertain variable.

Table 1 Example of graded lexicographic ordering with $n_\xi = 3$

$ j $	j	j'
0	(0, 0, 0)	0
1	(1, 0, 0)	1
—	(0, 1, 0)	2
—	(0, 0, 1)	3
2	(2, 0, 0)	4
—	(1, 1, 0)	5
—	(1, 0, 1)	6
—	(0, 2, 0)	7
—	(0, 1, 1)	8
—	(0, 0, 2)	9
3	(3, 0, 0)	10
—	(2, 1, 1)	11
$\vdots$	$\vdots$	$\vdots$

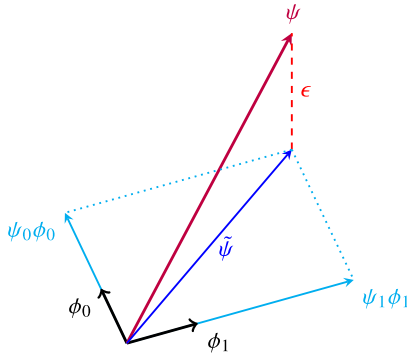


Fig. 1 GALERKIN projection of random variable ψ onto the space of two polynomial chaos basis functions ϕ_1 and ϕ_2 .

E. GALERKIN Projection of Stochastic Systems

As in [40], consider a random function $s^*(t, \xi)$, with n_s components $s_i^*(t, \xi)$ that are parameterized by ξ for $i \in \{1, \dots, n_s\}$. These components can be approximated using gPCE as follows:

$$s_i^*(t, \xi) \approx \tilde{s}_i(t, \xi) := \sum_{|j|=0}^p \underline{s}_{i,j}(t) \phi_j(\xi) \quad (33)$$

where $\underline{s}$ are the gPC expansion coefficients. For compactness, we define

$$\tilde{s}(t, \xi) := [\tilde{s}_1(t, \xi) \ \tilde{s}_2(t, \xi) \ \dots \ \tilde{s}_{n_s}(t, \xi)]^\top \in \mathbb{R}^{n_s \times 1} \quad (34a)$$

If $\tilde{s}$ was the state vector for a dynamical system, for example, then $\tilde{s}_1, \dots, \tilde{s}_{n_s}$ would be the individual components, and $s^*(t, \xi) \approx \tilde{s}(t, \xi)$. Next, we define

$$\underline{s}_i(t) := [\underline{s}_{i,0}(t) \ \underline{s}_{i,1}(t) \ \dots \ \underline{s}_{i,j'}(t)]^\top \in \mathbb{R}^{N_p \times 1} \quad (34b)$$

which is the column vector of N_p expansion coefficients for a given component i of $\tilde{s}$. Additionally, define

$$\phi(\xi) := [\phi_0(\xi) \ \phi_1(\xi) \ \dots \ \phi_{j'}(\xi)]^\top \in \mathbb{R}^{N_p \times 1} \quad (34c)$$

which is the column vector of N_p multivariate basis functions corresponding to each expansion coefficient. Finally, we group the n_s column vectors $\underline{s}_i$ to obtain

$$s_{PC}(t) := [\underline{s}_1(t) \ \dots \ \underline{s}_{n_s}(t)] \in \mathbb{R}^{N_p \times n_s} \quad (34d)$$

We can now represent the gPCE as

$$\tilde{s}(t, \xi) = s_{PC}^\top(t) \phi(\xi) \quad (35a)$$

$$= (\phi^\top(\xi) \otimes \mathbb{I}_{n_s}) \text{vec}(s_{PC}^\top(t)) \quad (35b)$$

$$=: \Phi_s^\top(\xi) S(t) \quad (35c)$$

where $\otimes$ is the KRONECKER product, and $\text{vec}(\cdot)$ is the vectorization of a matrix. Note that $\Phi_s^\top(\xi) \in \mathbb{R}^{n_s \times n_s N_p}$ and $S(t) \in \mathbb{R}^{n_s N_p \times 1}$.

F. GALERKIN Projection of the Uncertain Filter Equations

For clarity, let us use $(\cdot)^*$ to denote the original uncertain filter equation variables [Eq. (25)]. Substituting the gPCEs of $\hat{x}_A^*$ and $\hat{z}^*$ into the uncertain filter equations gives

$$\begin{aligned} \Phi_{\hat{x}_A}^\top(\xi) \dot{\hat{X}}_A(t) &= A_A(\xi) \Phi_{\hat{x}_A}^\top(\xi) \hat{X}_A(t) + B_A(\xi) u(t) \\ &\quad + K(y_A(t) - C_A \Phi_{\hat{x}_A}^\top(\xi) \hat{X}_A(t)) \end{aligned} \quad (36a)$$

$$\Phi_{\hat{z}}^\top(\xi) \dot{\hat{Z}}(t) = L \Phi_{\hat{x}_A}^\top(\xi) \hat{X}_A(t) \quad (36b)$$

Next, premultiply the filter equation by $\Phi_{\hat{x}_A}(\xi)$ and the estimate equation by $\Phi_{\hat{z}}(\xi)$ to get

$$\begin{aligned} \Phi_{\hat{x}_A}(\xi) \Phi_{\hat{x}_A}^\top(\xi) \dot{\hat{X}}_A(t) &= \Phi_{\hat{x}_A}(\xi) A_A(\xi) \Phi_{\hat{x}_A}^\top(\xi) \hat{X}_A(t) + \Phi_{\hat{x}_A}(\xi) B_A(\xi) u(t) \\ &\quad + \Phi_{\hat{x}_A}(\xi) K(y_A(t) - C_A \Phi_{\hat{x}_A}^\top(\xi) \hat{X}_A(t)) \end{aligned} \quad (37a)$$

$$\Phi_{\hat{z}}(\xi) \Phi_{\hat{z}}^\top(\xi) \dot{\hat{Z}}(t) = \Phi_{\hat{z}}(\xi) L \Phi_{\hat{x}_A}^\top(\xi) \hat{X}_A(t) \quad (37b)$$

Take the expectation of both equations and let $M_{\hat{x}_A} = \mathbb{E}_\xi\{\Phi_{\hat{x}_A}(\xi) \Phi_{\hat{x}_A}^\top(\xi)\}$ and $M_{\hat{z}} = \mathbb{E}_\xi\{\Phi_{\hat{z}}(\xi) \Phi_{\hat{z}}^\top(\xi)\}$ be the normalization matrices. By Eqs. (30) and (35), $M_{\hat{x}_A}$ and $M_{\hat{z}}$ are diagonal and full rank. Therefore

$$M_{\hat{x}_A} \dot{\hat{X}}_A(t) = A_A \hat{X}_A(t) + B_A u(t) + K y_A(t) - K C_A \hat{X}_A(t) \quad (38a)$$

$$M_{\hat{z}} \dot{\hat{Z}}(t) = L \hat{X}_A(t) \quad (38b)$$

where

$$A_A = \mathbb{E}_\xi\{\Phi_{\hat{x}_A}(\xi) A_A(\xi) \Phi_{\hat{x}_A}^\top(\xi)\} \in \mathbb{R}^{n_i N_p \times n_i N_p} \quad (39a)$$

$$B_A = \mathbb{E}_\xi\{\Phi_{\hat{x}_A}(\xi) B_A(\xi)\} \in \mathbb{R}^{n_i N_p \times n_u} \quad (39b)$$

$$K = \mathbb{E}_\xi\{\Phi_{\hat{x}_A}(\xi) K\} \in \mathbb{R}^{n_i N_p \times n_y} \quad (39c)$$

$$K C_A = \mathbb{E}_\xi\{\Phi_{\hat{x}_A}(\xi) K C_A \Phi_{\hat{x}_A}^\top(\xi)\} \in \mathbb{R}^{n_i N_p \times n_i N_p} \quad (39d)$$

$$L = \mathbb{E}_\xi\{\Phi_{\hat{z}}(\xi) L \Phi_{\hat{x}_A}^\top(\xi)\} \in \mathbb{R}^{n_z N_p \times n_i N_p} \quad (39e)$$

Finally,

$$\dot{\hat{X}}_A(t) = M_{\hat{x}_A}^{-1} (A_A \hat{X}_A(t) + B_A u(t) + K y_A(t) - K C_A \hat{X}_A(t)) \quad (40a)$$

$$\dot{\hat{Z}}(t) = M_{\hat{z}}^{-1} L \hat{X}_A(t) \quad (40b)$$

where $\hat{X}_A(t) \in \mathbb{R}^{n_i N_p}$. These gPC filter equations are no longer a function of ξ and are thus deterministic dynamics of expansion coefficients. The outputs are useful for analyzing the statistics of the behavior of the original uncertain system.

IV. Uncertainty Quantification and Sensitivity Analysis

This section describes the extraction of statistical moments and the computation of pointwise and cumulative sensitivity indices from the generalized polynomial chaos (gPC) filter formulation.

A. Statistical Information from the gPC Filter Equations

From the gPCE of Eq. (25), we now have an analytical representation [37] (Chap. 5) of the filter equations in terms of ξ . From this analytical representation, we can directly extract the statistics of the outputs. Recall from Sec. III.E that

$$\hat{z}^*(t, \xi) \approx \Phi_z^\top(\xi) \hat{Z}(t) \quad (41)$$

The mean of $\hat{z}^*(t, \xi)$ over ξ can thus be approximated as

$$\begin{aligned} \mu_{z^*}(t) &:= \mathbb{E}_\xi \{ \hat{z}^*(t, \xi) \} \approx \mathbb{E}_\xi \{ \Phi_z^\top(\xi) \hat{Z}(t) \} \\ &= \int_\Omega \left(\Phi_z^\top(\xi) \hat{Z}(t) \right) \varphi(\xi) d\xi = \hat{Z}_0(t) \end{aligned} \quad (42)$$

Similarly, its covariance, $\text{Cov}(\hat{z}^*(t, \xi))$, can be approximated from

$$\begin{aligned} \text{Cov}(\hat{z}^*(t, \xi)) &:= \mathbb{E}_\xi \left\{ \left(\hat{z}^*(t, \xi) - \mu_{z^*}(t) \right) \left(\hat{z}^*(t, \xi) - \mu_{z^*}(t) \right)^\top \right\} \\ &\approx \mathbb{E}_\xi \left\{ \left(\Phi_z^\top(\xi) \hat{Z}(t) - \hat{Z}_0(t) \right) \left(\Phi_z^\top(\xi) \hat{Z}(t) - \hat{Z}_0(t) \right)^\top \right\} \end{aligned} \quad (43a)$$

$$\approx \mathbb{E}_\xi \left\{ \left(\Phi_z^\top(\xi) \hat{Z}(t) - \hat{Z}_0(t) \right) \left(\Phi_z^\top(\xi) \hat{Z}(t) - \hat{Z}_0(t) \right)^\top \right\} \quad (43b)$$

$$= \mathbb{E}_\xi \left\{ \Phi_z^\top(\xi) \hat{Z}(t) \hat{Z}^\top(t) \Phi_z(\xi) \right\} - \hat{Z}_0(t) \hat{Z}_0^\top(t) \quad (43c)$$

$$= M_z \hat{Z}(t) \hat{Z}^\top(t) - \hat{Z}_0(t) \hat{Z}_0^\top(t) \quad (43d)$$

Thus, the total variance of $\hat{z}^*(t, \xi)$ is $\sigma^2(\hat{z}^*(t, \xi)) = \text{tr}(\text{Cov}(\hat{z}^*(t, \xi)))$. Other statistical information, such as coskewness and cokurtosis, can be determined similarly [41].

B. Pointwise Sensitivity Analysis

Recall the tuple j defined in Sec. III.D. Now, define $\mathcal{J}_a$ as the set of j such that only the entries $a = \{i_1, \dots, i_s\}$ are nonzero:

$$\mathcal{J}_a = \{j | j_k = 0 \iff k \notin a\} \quad (44)$$

From Table 1, each tuple j corresponds to a single-index j' (i.e., $j \hat{=} j'$) according to the GLO scheme. Thus, the pointwise-in-time gPC-based SOBOL'-SUDRET sensitivity indices [42] are

$$S_{U_a}(t) = \sum_{\substack{j \in \mathcal{J}_a \\ j \hat{=} j'}} M_{\hat{z}_{j'}, j'} Z_{j'}(t) Z_{j'}^\top(t) / \sigma^2(\hat{z}^*(t, \xi)) \quad (45)$$

The sensitivity indices of a single parameter (e.g., S_{U_1} , S_{U_2} , etc.) are referred to as *main effects*. Indices involving multiple parameters (e.g., $S_{U_{1,2}}$, $S_{U_{2,3}}$, etc.) are known as *interaction effects*. Let $\#a$ be the cardinality of a . Note that a direct result of the construction of the gPC basis functions [Eq. (28)] is that $p \geq \#a$. In other words, the number of interactions (pairwise, triplewise, etc.) able to be evaluated is dictated by the expansion order. This can be seen in the example GLO scheme shown in Table 1. Define S_U as the set of all main and interaction effects:

$$S_U = \{S_{U_a}(t) | a = \{i_1, \dots, i_s\} \subseteq \{1, \dots, n_\xi\}\} \quad (46)$$

The *total effect* (e.g., $S_{U_1^{\text{tot}}}$, $S_{U_2^{\text{tot}}}$) is the sum of the main effect and all interaction effects associated with a given parameter:

$$S_{U_b^{\text{tot}}}(t) = \sum_{\{S_{U_a}(t) \in S_U | b \in a\}} S_{U_a}(t) \quad (47)$$

These pointwise sensitivities are useful for conducting SA close to a specific time of interest; however, they ignore temporal correlations,

which may be significant in time-dependent processes such as wind estimation [43]. Alternatively, evaluating cumulative effects can provide additional insight as to which parameters are most significant.

C. Cumulative Sensitivity Analysis

The cumulative-over-time gPC-based SOBOL'-SUDRET-Alexanderian sensitivity indices [43] are

$$S_{A_a}(T) = \frac{\sum_{\substack{j \in \mathcal{J}_a \\ j \hat{=} j'}} \int_0^T M_{\hat{z}_{j'}, j'} Z_{j'}(t) Z_{j'}^\top(t) dt}{\int_0^T \sigma^2(\hat{z}^*(t, \xi)) dt} \quad (48)$$

By examining Eqs. (45) and (48), we can see that the cumulative indices include an integral effect in the numerator and denominator that is not present in the pointwise indices. In this way, a parameter that has a small yet constant effect on the output can be compared to a parameter that has a larger but momentary effect. Methods such as windowing [44] can also be used to compute limited-memory sensitivities, but they are not pursued in this study.

As before, define S_A as the set of all main and interaction cumulative indices:

$$S_A = \{S_{A_a}(T) | a = \{i_1, \dots, i_s\} \subseteq \{1, \dots, n_\xi\}\} \quad (49)$$

Thus

$$S_{A_b^{\text{tot}}}(T) = \sum_{\{S_{A_a}(T) \in S_A | b \in a\}} S_{A_a}(T) \quad (50)$$

For completeness, summing all the main and interaction effects, pointwise or cumulative, recovers the total variance

$$\sigma^2(\hat{z}^*(t, \xi)) = \sum_{\{S_{U_a}(t) \in S_U\}} S_{U_a}(t) = \sum_{\{S_{A_a}(T) \in S_A\}} S_{A_a}(T) \quad (51)$$

V. Sensitivity-Informed Variable Reduction

This section describes a sensitivity-informed reduction of the generalized polynomial chaos expansion that limits the number of retained random variables and interaction terms to reduce computational cost.

Full characterization of all interaction effects among the random variables would require $p \geq n_\xi$. Recalling Eq. (32), for a large number of variables, the computation of gPC quickly becomes intractable. Alternatively, we can use the information obtained from the sensitivity indices to systematically reduce the number of random variables that are carried from one gPCE order to the next. This avoids unnecessary computation of insignificant interaction effects or higher-order expansions for variables well described by a low-order expansion. The choice of variables to retain may be based on a predefined threshold [44], convergence to a final value [42], an ordered ranking that is limited based on computation time, or another criterion. Then, a first-order gPCE is performed to obtain an initial estimate of the main effect indices to decide which variable to retain. Next, a second-order expansion involving the retained variables is conducted to refine their main effect indices and obtain pairwise interactions. This process is repeated until all criteria have been satisfied. We call this approach *reduced generalized polynomial chaos* (gPCr). Although variable reduction is not necessary for gPC UQ [45] or SA, it will be shown in Sec. VII that the number of expansion factors, and therefore the computational burden, can be greatly reduced while achieving accurate UQ results.

Let $i n_\xi \leq n_\xi$ be the number of random variables retained in the i th gPCE. Following the process described in Sec. III.D, the multivariate basis functions are constructed using only the $i n_\xi$ variables for each expansion order:

$$\phi_j'(\xi) = \phi_{j_1}(\xi_1) \dots \phi_{j_{i n_\xi}}(\xi_{i n_\xi}), \quad 0 \leq |j| \leq p, \quad 0 \leq i \leq p \quad (52)$$

The remaining steps to achieve a deterministic gPC system are unchanged. The number of expansion factors resulting from this reduction method N_p' is given by

$$N'_p = 1 + \sum_{i=1}^p \left(\frac{(i n_\xi + i)!}{(i n_\xi)! (i)!} - \frac{(i n_\xi + i - 1)!}{(i n_\xi)! (i - 1)!} \right) \leq N_p \quad (53)$$

An example of this variable reduction method is provided in Algorithm 1.

Algorithm 1 Sensitivity-Informed Variable Reduction

```

1: Retention criterion 1:  $\max(S_{\square}(T)) > C_1$ 
2: Retention criterion 2:  $\Delta(\max(S_{\square}(T))) > C_2$ 
3: procedure PERFORM GPCE USING  $n_\xi$  GERMS.
4:   Expand system using all  $n_\xi$  germs.
5:   Evaluate all main effect sensitivity indices using retention criterion 1.
6:   Retain  $n_\xi$  germs associated with retention criterion 1.
7: end procedure
8: for  $2 \leq i \leq p$  do
9:   if  $i n_\xi > 0$  then
10:    Expand system using all  $i n_\xi$  germs.
11:    Evaluate new sensitivity indices using retention criterion 1.
12:    Evaluate updated sensitivity indices using retention criterion 2.
13:    Retain  $(i+1)n_\xi$  germs associated with retention criterion 1 and 2.
14:   else
15:     break
16:   end if
17: end for

```

VI. Multivariate Uncertainty Quantification and Sensitivity Analysis from Flight-Test Data

A. Research Aircraft

The research aircraft used in this study is the My Twin Dream (MTD), shown in Fig. 2, manufactured by MyFlyDream. It is a radio-controlled foam aircraft with counter-rotating twin electric motors and APC propellers measuring 10 in. in diameter with a 6 in. pitch. The aircraft was instrumented with a CubePilot Cube Orange flight computer running PX4 firmware. The sensors onboard the aircraft include triple-redundant accelerometers and gyroscopes, two magnetometers, a real-time kinematic GPS receiver, and a vaned air data unit (ADU) for wind estimate validation. The MTD was chosen for its simple construction, propeller location (to accommodate the air data boom), and endurance of approximately 25 min. The MTD physical properties are listed in Table 2.

The aerodynamic force and moment coefficients in Eq. (10) are taken from reproduced in Table A1 in Appendix A, along with their associated mean μ , standard deviation σ , and coefficient of variation (CV) [46]. The propulsion coefficient comes from a model based on the wind tunnel data from [23].

The accelerometers, gyroscopes, magnetometers, and GPS feed into an extended KALMAN filter as part of the PX4 firmware, from which we used position $\mathbf{X}$, attitude $\boldsymbol{\Theta}$, and angular velocity $\boldsymbol{\omega}$, which were provided along with their respective state estimate error variances [47].

The wind estimates were compared with reconstructed wind data from a vaned ADU mounted on the nose of the aircraft (Fig. 2). This sensor was developed, manufactured, and calibrated by the Nonlinear Systems Laboratory at the Virginia Tech [26,27]. It consists of two 3D-printed vanes attached to magnetic rotary encoders and a 3D-printed Kiel probe connected to a MS5525DSO[†] pressure sensor. The pulse-width modulation rotary encoders are read at a sample rate of 200 Hz by a microcontroller that communicates with the flight computer over the Controller Area Network (CAN) bus via a custom PX4 driver. The pressure sensor

is natively supported by PX4 and configured to log calibrated airspeed data at 10 Hz.

Let V , α , and β_f be the airspeed, angle of attack, and flank angle reported by the ADU, respectively. By using GPS velocity measurements $\mathbf{v}_i$ (accuracy of 0.05 m/s), autopilot attitude estimates (with quaternion estimate standard deviations $\approx 10^{-3}$), and angular velocity measurements from the calibrated gyroscope (noise and bias removed), the wind velocity may be reconstructed as

$$\mathbf{w} = \mathbf{v}_i - \mathbf{R}_{IB}(\mathbf{v}_{\text{ADU}} - \boldsymbol{\omega} \times \mathbf{r}_{\text{ADU}}) \quad (54)$$

where

$$\mathbf{v}_{\text{ADU}} = \mathbf{R}_{\text{BW}}(\alpha, \beta) \mathbf{e}_1 V_r$$

Here, $\mathbf{v}_{\text{ADU}}$ is the air-relative velocity at the geometric center of the ADU vanes, whose position in the body frame is denoted $\mathbf{r}_{\text{ADU}}$. The rotation matrix $\mathbf{R}_{\text{BW}}(\alpha, \beta) = e^{-[e_2 \times] \alpha} e^{[e_3 \times] \beta}$, which maps free vectors from the wind frame to the body frame, is parameterized by the measured angle of attack α and the sideslip angle $\beta = \tan^{-1}(\tan(\beta_f) \cos(\alpha))$. The accuracy of the reconstructed wind data can be characterized by propagating the measurement uncertainty through Eq. (54). Because of the low error of the constituent measurements, the reconstructed wind is treated as a truth source for wind estimate validation. This ADU was validated in the open-jet wind tunnel of the Aerospace and Ocean Engineering Department at Virginia Tech, for both steady-state readings and dynamic response. The airspeed-dependent cutoff frequencies of the vanes were found to be greater than 16 Hz, well above the fastest rigid-body modes, which determine an upper limit on the dynamic response of model-based wind estimators.

B. Flight Test

A flight-test campaign was conducted on 28 September 2022 at the Kentland Experimental Aerial Systems airfield of Virginia Tech, using the UAS described in Sec. VI.A. Data were gathered during four passes over the ground control station (runs 1–4, as shown in Fig. 3) at 76 m (250 ft) above ground level when the wind conditions at ground level were gusty. Gusty conditions were chosen because the goal was to achieve accurate wind estimates under any condition in which the UAS is able to fly. Because we used an LTI model, we restricted the maneuver to nominally steady flight, namely, straight-and-level unaccelerated flight.

C. Methodology

The first step of conducting UQ and SA is to determine how the uncertainty in the parameters of the nonlinear model, given in Table A1 (Appendix A), maps to uncertainty in the entries of the linear system matrices. This was done analytically according to Sec. III.B. The resulting statistics for the $\mathbf{A}(\boldsymbol{\chi})$ matrix entries are presented in Table 3. Note that only the entries affected by changes in aerodynamic parameters (i.e., dynamic states and derivatives) are given. Next, to create the nominal Luenberger filter (denoted with the subscript “nom”), the filter gain matrix $\mathbf{K}_{\text{nom}}$ was found using a deterministic method based on the nominal system, where $\mathbf{A}_{\text{nom}} = \mathbb{E}_{\boldsymbol{\xi}}\{\mathbf{A}(\boldsymbol{\xi})\}$. For the results shown here, H_∞ filtering, as presented in [26,27], was used. Alternatively, another method such as KALMAN–BUCY filtering could be used.

Next, we performed gPCE according to Sec. III.D. Although there is no limit to the number of entries that can be included, the computational burden of performing gPCE grows with N_p [Eq. (32)]. This is commonly referred to as the *curse of dimensionality*. Instead, we used the gPCr method described in Sec. V. Because the distribution of each stochastic entry is GAUSSIAN, the appropriate gPC polynomial basis is the family of HERMITE polynomials [37], given by

[†]Manufactured by TE Connectivity.



Fig. 2 MTD aircraft.

Table 2 MTD properties

Property	Symbols	Values	Units
Mass	m	3.311	kg
Mean aerodynamic chord	$\bar{c}$	0.254	m
Projected wingspan	b	1.800	m
Wing planform area	S	0.457	m ²
Roll moment of inertia	I_{xx}	0.319	kg·m ²
Pitch moment of inertia	I_{yy}	0.267	kg·m ²
Yaw moment of inertia	I_{zz}	0.471	kg·m ²
Product of inertia	I_{xz}	0.024	kg·m ²
Product of inertia	I_{xy}, I_{yz}	≈ 0	kg·m ²

$$\text{He}_n(\xi) = (-1)^n e^{\frac{\xi^2}{2}} \frac{d^n}{d\xi^n} e^{-\frac{\xi^2}{2}} \quad (55)$$

The first five HERMITE polynomials are

$$\begin{aligned} \text{He}_0 &= 1, \text{He}_1 = \xi, \text{He}_2 = \xi^2 - 1, \text{He}_3 = \xi^3 - 3\xi, \\ \text{He}_4 &= \xi^4 - 6\xi^2 + 3 \end{aligned} \quad (56)$$

with the corresponding probability density function

$$\varphi(\xi) = \frac{1}{\sqrt{2\pi}} e^{-\frac{\xi^2}{2}}, \text{ with support } \Omega = (-\infty, \infty) \quad (57)$$

Recalling Sec. III.D, we are not constrained to an assumption of normality to apply gPC. However, given the nature of the uncertainties being considered, GAUSS–HERMITE polynomials are appropriate. MC simulations were conducted, and gPC was performed to compare the computational cost and accuracy of the two methods. For the first-order expansion ($p = 1$) of gPCr, we retained all 14 entries ($^1n_\xi = 14$) given in Table 3. Subsequent expansions only retain entries that meet a certain criterion, which is described in Sec. VII. For the MC simulations, the filter gain matrix $\mathbf{K}_{\text{nom}}$ did not change. Only the state matrix changed to capture the uncertainty between the “actual” aircraft dynamics and the “assumed” dynamics used in the filter. The variation in the family of estimates $\{\hat{w}_N, \hat{w}_E, \hat{w}_D\}$ in Sec. VII is the result of the uncertainty in the state matrix, $\mathbf{A}(\xi)$, which was ignored in the design of the filter. Thus, the form of the filter equations used for the MC simulations was

$$\dot{\hat{\mathbf{x}}}_A(t, \xi) = \mathbf{A}_A(\xi) \hat{\mathbf{x}}_A(t, \xi) + \mathbf{B}_A \mathbf{u}(t) + \mathbf{K}_{\text{nom}} (\mathbf{y}_A(t) - \mathbf{C}_A \hat{\mathbf{x}}_A(t, \xi)) \quad (58a)$$

$$\hat{\mathbf{z}}(t, \xi) = \mathbf{L} \hat{\mathbf{x}}_A(t, \xi) \quad (58b)$$

Because MC results are used as a baseline for evaluating the gPC results, we must ensure that the MC simulations have converged,

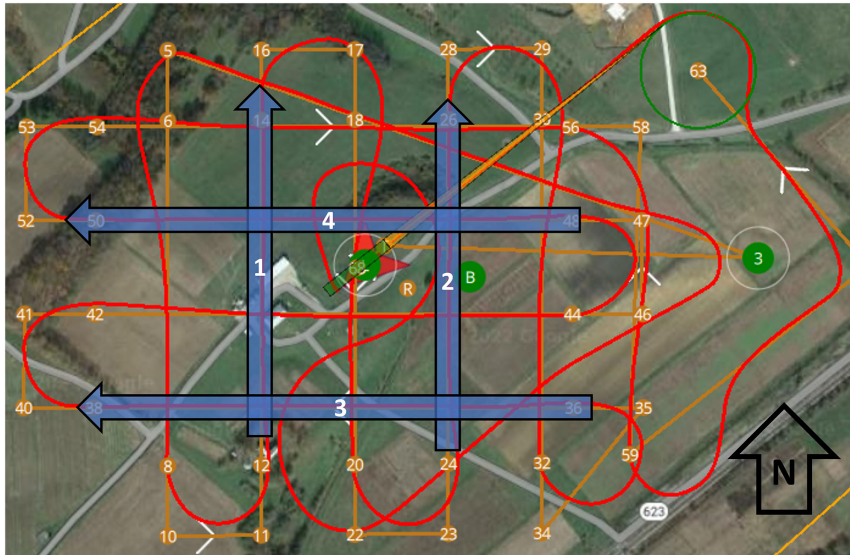


Fig. 3 Runs 1–4 from the 28 September 2022 flight test campaign; the prevailing wind was out of the northwest during the test campaign.

Table 3 Uncertain entries of linear $A(\chi)$ matrix

Axis of motion	Derivatives	State	Matrix	Variable	Mean	Standard deviation	CV
	(row)	(col.)	entry	no.	(μ)	(σ)	($\sigma/ \mu $)
Longitudinal	$\dot{u}_r(7)$	$u_r(7)$	$A_{7,7}$	1	-0.197	0.090	0.46
	$\dot{w}_r(9)$	$u_r(7)$	$A_{9,7}$	2	-0.932	0.276	0.28
	$\dot{q}(11)$	$u_r(7)$	$A_{11,7}$	3	+0.015	0.066	4.12
		$w_r(9)$	$A_{11,9}$	7	-2.416	0.147	0.06
		$q(11)$	$A_{11,11}$	11	-9.684	0.233	0.02
Lateral-directional	$\dot{v}_r(8)$	$v_r(8)$	$A_{8,8}$	4	-0.708	0.200	0.28
		$p(10)$	$A_{8,10}$	8	+0.474	0.207	0.44
		$r(12)$	$A_{8,12}$	12	-19.64	0.847	0.04
	$\dot{p}(10)$	$v_r(8)$	$A_{10,8}$	5	-0.997	0.133	0.13
		$p(10)$	$A_{10,10}$	9	-11.27	1.194	0.11
		$r(12)$	$A_{10,12}$	13	-0.177	0.043	0.24
	$\dot{r}(12)$	$v_r(8)$	$A_{12,8}$	6	+1.767	0.440	0.25
		$p(10)$	$A_{12,10}$	10	-0.574	0.061	0.10
		$r(12)$	$A_{12,12}$	14	-2.352	0.573	0.24

meaning that a sufficient number of simulations have been conducted to accurately capture the uncertainty in the output. Using the variance in the filter estimates as the quantity of interest, for the results presented, we used a doubling scheme for the number of MC simulations N_{MC} and continued until the change in variance $\Delta\sigma_{MC}^2$ from N_{MC} to $2N_{MC}$ was less than a desired percentage (1% for the results shown here). Because MC converges at a rate $O(N_{MC}^{-1/2})$ [48], this double-and-check scheme was seen as a favorable method for establishing the MC baseline.

VII. Results

The simulations presented in this section were performed in MATLAB R2023b on a laptop with an AMD Ryzen 7 PRO 7840U CPU with eight physical cores and 64 GB of DDR5-6400 memory. The MC simulations were accomplished using MATLAB's `parfor` command,** and the gPC computations were accomplished analytically using symbolic integration along with `parfor`.

The results presented in this section are based on data from run 1 of the flight tests described in Sec. VI.B. The MC simulation convergence criterion $\Delta\sigma_{MC}^2$ was chosen as 1%, which required 255,000 simulations ($N_{MC} = 255,000$). Figure 4 shows all MC simulations of the nominal filter [Eq. (58)], allowing for uncertainty in all 14 entries of $A(\xi)$. As can be seen, the north component $\hat{w}_N$ is the most affected by these uncertainties; that is, it has the largest variance in estimates over time, followed by the east component $\hat{w}_E$. This is expected because for run 1, the aircraft flew a south-to-north course. Thus, the longitudinal dynamics, which have dominant uncertainties (as will be shown briefly), were used to resolve the north component of the wind. Similar results were seen for run 2. Conversely, on runs 3 and 4, the east component had the largest variance. The downward component $\hat{w}_D$ was much less affected. For reference, the actual wind values (w_N, w_E, w_D) measured by the ADU, as discussed in Sec. VI.A, are also shown.

For gPCr, two criteria were used to determine which variables to retain. The first criterion is that the maximum value of its cumulative main or any interaction index is greater than 0.03: $\max(S_{A_n}(T)) > 0.03$. The second criterion is that, for indices greater than 0.03, the value of the index changes from one expansion order to the next. Figures 5 and 6 show the time history of the cumulative and pointwise indices, respectively. Because we are interested in determining which key parameters contribute to

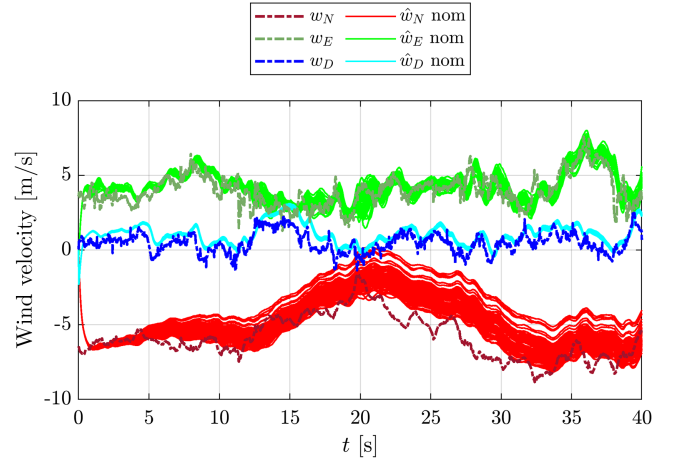


Fig. 4 MC simulations of the nominal (nom) filter with uncertainty in the longitudinal entries of $A(\chi)$.

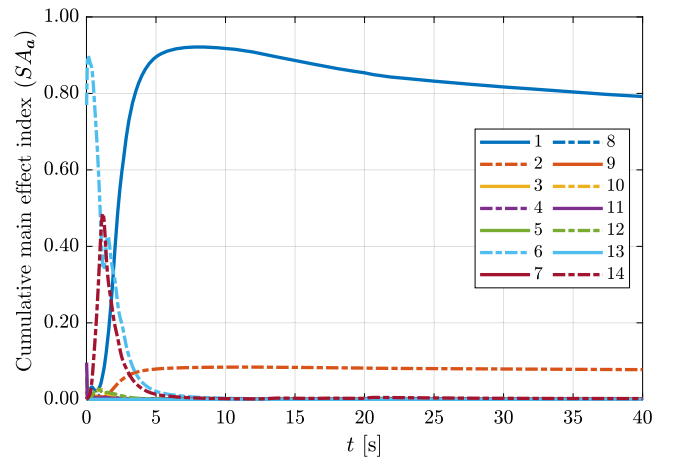


Fig. 5 Cumulative main effect sensitivity indices computed using first-order gPCr with $n_\xi = 14$ for $\hat{w}_N$.

the wind estimate variance over time, we focus our analysis on the cumulative indices. As an alternative, one may find it useful to use the pointwise results to analyze local variations, such as those shown around the 20 s mark in Fig. 6.

**<https://www.mathworks.com/products/parallel-computing.html>.

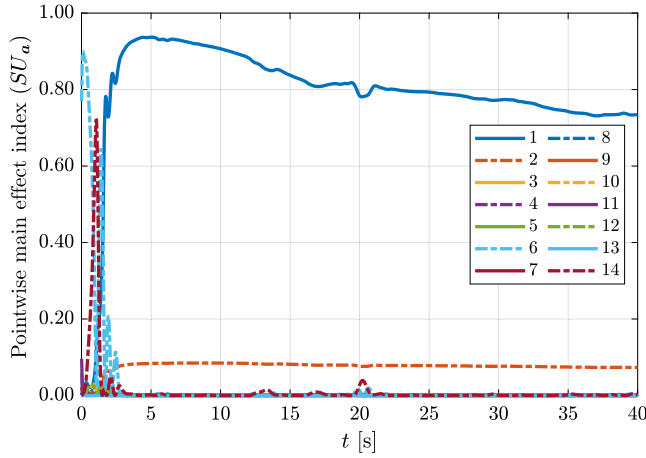


Fig. 6 Pointwise main effect sensitivity indices computed using first-order gPC with $n_g = 14$ for $\hat{w}_N$.

From Fig. 5, we can clearly see four of the five variables (1, 2, 6, and 14) that satisfied the first criterion. Variable 11 also satisfied the criterion close to $t = 0$, but it decayed quickly. Nevertheless, we retained this term for the next gPCr expansion. Table 4 shows the results for the first-order and second-order gPCr expansions. In the first row ($p = 1$), five variables satisfied the first criterion. Moving to the second row ($p = 2$), we are now able to characterize interaction effects among two variables and refine the main effect indices. Based on the first criterion, $\max(S_{A_a}(T)) > 0.03$, the interaction effects $S_{A_{1,2}}$ and $S_{A_{6,14}}$ are not significant. However, the indices S_{A_1} , S_{A_2} , S_{A_6} , and $S_{A_{14}}$ have all changed; therefore, we retained these variables for the next expansion. On the other hand, $S_{A_{11}}$ did not change from $p = 1$ to $p = 2$ and is also not part of a significant interaction effect; therefore, we did not include this variable in the next expansion. The final row ($p = 3$) of Table 4 shows that both criteria were satisfied. Therefore, there is no need to perform an additional expansion.

Table 5 presents the times for the MC and gPCr to compute the total variance, as well as the average and maximum error in variance between the gPCr approximations and the MC baseline data for $\hat{w}_N$. Error is defined as $\epsilon(\sigma^2) = \sigma_{MC}^2 - \sigma_{PCE}^2$. It can be seen that a second-order gPCr accurately captures the variance in

$\hat{w}_N$ due to uncertainty in $A(\xi)$ while achieving a 38% reduction in computation time. The third-order gPCE requires significantly more computation time for marginal improvements. Figure 7 shows that all three tested orders of gPCE can accurately track the MC mean wind. This is expected because, recalling Eq. (42), the mean wind is given by the zeroth gPC expansion coefficient of each wind component.

Figure 8, in contrast, shows the improvement in variance tracking as the gPCE order increases. The green curve, representing the second-order gPCE, is obscured by the blue curve because their values are nearly identical. In general, the rate of convergence of gPCE orders is dictated by the compatibility of the random parameters with the chosen basis functions and whether the system depends smoothly on the random parameters [37] (Chap. 3). For the results presented here, the uncertainties in $A(\xi)$ are well represented by HERMITE polynomials, and the filter equations smoothly depend on $A(\xi)$. Thus, a second-order gPCr expansion is sufficient for characterizing the total variance.

Table 6, however, shows the true advantage of using gPCr over MC. Although a second-order gPCr expansion may be sufficient to simply characterize the wind estimate variance, according to the chosen criteria, a third-order expansion is necessary to conduct SA.

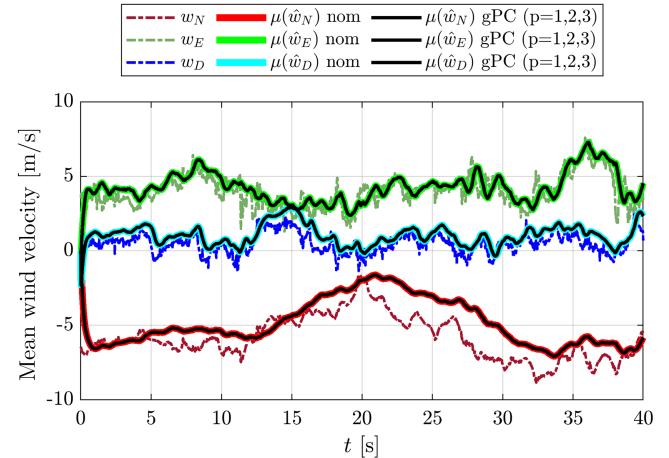


Fig. 7 Mean wind velocities from MC of the nominal (nom) filter and single runs of gPCr filters of different orders.

Table 4 Cumulative sensitivity indices computed using gPCr with $n_g = 14$ for $\hat{w}_N$

p	n_g	N'_p	$\max(S_{A_a}(T))$									
			S_{A_1}	S_{A_2}	S_{A_6}	$S_{A_{11}}$	$S_{A_{14}}$	$S_{A_{\bullet}}$	$S_{A_{1,2}}$	$S_{A_{6,14}}$	$S_{A_{\bullet,\bullet}}$	$S_{A_{\bullet,\bullet,\bullet}}$
1	14	15	0.922	0.084	0.898	0.096	0.486	<0.03	N/A	N/A	N/A	N/A
2	5	30	0.899	0.099	0.901	0.096	0.484	^a	0.020	0.026	~0	N/A
3	4	50	0.899	0.099	0.901	^a	0.484	^a	0.020	0.026	~0	~0

^aVariables not included in gPC expansion.

Note: $\bullet$ All other indices that are not enumerated.

Table 5 Variance computation time and error for gPCr of increasing orders with $n_g = 14$

Method	Runs	Computation time [mm:ss]				$\epsilon(\sigma^2(\hat{w}_N))$			
		gPCr	Simulation	Total		Average		Maximum	
						In m^2/s^2	In %	In m^2/s^2	In %
Monte Carlo	255,000	N/A	10:57	10:57		—	—	—	—
gPCr ($p = 1, n_g = 14$)	1	01:53	00:02	01:55		0.051	12.6	0.106	16.0
gPCr ($p = 2, n_g = 5$)	1	06:43	00:05	06:48		0.007	1.9	0.014	2.2
gPCr ($p = 3, n_g = 4$)	1	21:39	00:15	21:54		0.003	0.7	0.007	1.0

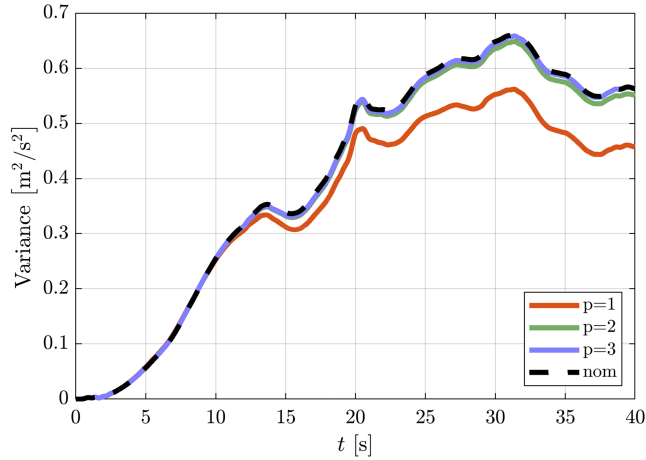


Fig. 8 Variance in $\hat{w}_N$ calculated from MC of the nominal (nom) filter and single runs of gPCr filters of different orders.

Table 6 Computation time and operations for cumulative sensitivity indices ($S_{A_\theta}(T)$) using gPCr, full gPC, and MC for $\hat{w}_N$

p	$i n_\xi$	Computation time [h:mm]			Operations ^a		
		Reduced gPC	Full gPC	MC	N'_p	N_p	N_{MC}
1	14	0:02	0:02	2:58	14	14	$\sim 4 \times 10^6$
2	5	0:07	1:52	4:44	30	120	$\sim 7 \times 10^6$
3	4	0:22	67:40	5:28	50	680	$\sim 8 \times 10^6$

^a N_p and N'_p are not equivalent to N_{MC} , but they are provided for reference.

Using MC to compute the main effect indices requires $N_{MC}(i n_\xi + 2)$ simulations [43], as shown in the first row of Table 6. Computing the interaction effects requires an additional $N_{MC} \binom{i n_\xi}{i}$ simulations, as shown in the second and third rows. These indices are provided by gPCr at no additional cost. Finally, the full gPC becomes prohibitively computationally expensive after a second-order expansion, further justifying a principled variable reduction method.

VIII. Conclusions

This paper presents the theory and process for obtaining a deterministic system of equations from a stochastic system using gPC, along with a systematic approach for reducing expansion variables (gPCr). The concepts were applied to the model-based wind estimation problem using measurements obtained from an unmanned aircraft. The aim was to obtain a measure of the precision of these wind estimates over time, given the known uncertainties in the model parameters. Additionally, gPCr was used to identify the critical sources of uncertainty via SA. For the multivariate case presented here, gPCr showed computational advantages over MC sampling-based methods. In practice, gPCr can be computed offline, with the resulting system of equations applied in real-time applications, such as in synthetic air data systems or path planning algorithms, which is not possible using MC. The deterministic system that results from gPC/gPCr can be used for a number of control and estimation [49] schemes. The methods presented here are extensible to nonlinear systems [28] and interdependent uncertainties [50,51]; therefore, the applications of gPC are manifold.

Appendix A: Nonlinear Model Parameters

Table A1 Nonlinear model parameters

Axis of motion	Coefficient	Parameter (θ_i)	Mean ($\mu(\theta_i)$)	Standard deviation ($\sigma(\theta_i)$)	CV ($\frac{\sigma}{\mu}$)
Longitudinal	C_X	C_{X_0}	+0.009	0.026	2.89
		C_{X_a}	+0.282	0.001	0.01
		$C_{X_{a^2}}$	+3.173	0.119	0.04
		$C_{X_{\delta_a}}$	+0.051	0.025	0.49
	C_Z	C_{Z_0}	-0.255	0.074	0.29
		C_{Z_a}	-4.436	0.015	0.00
		C_{Z_q}	-12.54	0.030	0.00
	C_m	C_{m_0}	+0.008	0.006	0.75
		C_{m_a}	-0.444	0.027	0.06
		C_{m_q}	-14.02	0.336	0.02
		$C_{m_{\delta_a}}$	-0.415	0.058	0.14
Lateral-directional	C_Y	C_{Y_β}	-0.410	0.115	0.28
		C_{Y_p}	+0.221	0.133	0.60
		C_{Y_r}	+0.230	0.547	2.38
		$C_{Y_{\delta_a}}$	+0.118	0.011	0.09
		$C_{Y_{\delta_r}}$	+0.136	0.030	0.22
	C_l	C_{l_β}	-0.035	0.004	0.11
		C_{l_p}	-0.386	0.041	0.11
		$C_{l_{\delta_a}}$	-0.137	0.010	0.07
	C_n	C_{n_β}	+0.083	0.020	0.24
		C_{n_r}	-0.119	0.029	0.24
		$C_{n_{\delta_a}}$	+0.013	0.009	0.69
		$C_{n_{\delta_r}}$	-0.068	0.006	0.09

Appendix B: Resulting Normal Sum Equations

The following are normal sum theorem formulas for the model parameters described in Sec. III.B. The resulting longitudinal entries of $\mathbf{A}(\chi)$ are

$$\begin{aligned}\mu(A_{7,7}) &= 3.46\mu(C_T) + 3.46\mu(C_{X_0}) + 0.011\mu(C_{X_a}) + 0.043\mu(C_{X_{\delta_e}}) \\ \mu(A_{9,7}) &= 3.46\mu(C_{z_0}) + 0.011\mu(C_{z_a}) \\ \mu(A_{11,7}) &= 10.9\mu(C_{m_0}) + 0.035\mu(C_{m_a}) + 0.134\mu(C_{m_{\delta_e}}) \\ \mu(A_{11,9}) &= 0.071\mu(C_{m_0}) + 5.44\mu(C_{m_a}) + 0.001\mu(C_{m_{\delta_e}}) \\ \mu(A_{11,11}) &= 0.691\mu(C_{m_q}) \\ \sigma^2(A_{7,7}) &= (3.46\sigma(C_T))^2 + (3.46\sigma(C_{X_0}))^2 + (0.011\sigma(C_{X_a}))^2 \\ &\quad + (0.043\sigma(C_{X_{\delta_e}}))^2 \\ \sigma^2(A_{9,7}) &= (3.46\sigma(C_{z_0}))^2 + (0.011\sigma(C_{z_a}))^2 \\ \sigma^2(A_{11,7}) &= (10.9\sigma(C_{m_0}))^2 + (0.035\sigma(C_{m_a}))^2 + (0.134\sigma(C_{m_{\delta_e}}))^2 \\ \sigma^2(A_{11,9}) &= (0.071\sigma(C_{m_0}))^2 + (5.44\sigma(C_{m_a}))^2 + (0.001\sigma(C_{m_{\delta_e}}))^2 \\ \sigma^2(A_{11,11}) &= (0.691\sigma(C_{m_q}))^2\end{aligned}$$

Likewise, the lateral-directional entries of $\mathbf{A}(\chi)$ are

$$\begin{aligned}\mu(A_{8,8}) &= 1.73\mu(C_{Y_\beta}), & \sigma^2(A_{8,8}) &= (1.73\sigma(C_{Y_\beta}))^2 \\ \mu(A_{8,10}) &= 1.56\mu(C_{Y_p}), & \sigma^2(A_{8,10}) &= (1.56\sigma(C_{Y_p}))^2 \\ \mu(A_{8,12}) &= 1.56\mu(C_{Y_r}), & \sigma^2(A_{8,12}) &= (1.56\sigma(C_{Y_r}))^2 \\ \mu(A_{10,8}) &= 32.4\mu(C_{l_\beta}) + 1.65\mu(C_{n_\beta}), & \sigma^2(A_{10,8}) &= (32.4\sigma(C_{l_\beta}))^2 \\ & & & \quad + (1.65\sigma(C_{n_\beta}))^2 \\ \mu(A_{10,10}) &= 29.2\mu(C_{l_p}), & \sigma^2(A_{10,10}) &= (29.2\sigma(C_{l_p}))^2 \\ \mu(A_{10,12}) &= 1.49\mu(C_{n_r}), & \sigma^2(A_{10,12}) &= (1.49\sigma(C_{n_r}))^2 \\ \mu(A_{12,8}) &= 1.65\mu(C_{l_\beta}) + 22.0\mu(C_{n_\beta}), & \sigma^2(A_{12,8}) &= (1.65\sigma(C_{l_\beta}))^2 \\ & & & \quad + (22.0\sigma(C_{n_\beta}))^2 \\ \mu(A_{12,10}) &= 1.49\mu(C_{l_p}), & \sigma^2(A_{12,10}) &= (1.49\sigma(C_{l_p}))^2 \\ \mu(A_{12,12}) &= 19.8\mu(C_{n_r}), & \sigma^2(A_{12,12}) &= (19.8\sigma(C_{n_r}))^2\end{aligned}$$

Acknowledgments

The authors gratefully acknowledge sponsorship from NASA under grant no. 80NSSC20M0162, as well as support from the U.S. Air Force and Kevin T. Crofton of the Department of Aerospace and Ocean Engineering at Virginia Tech.

References

- [1] González-Rocha, J., De Wekker, S. F. J., Ross, S. D., and Woolsey, C. A., "Wind Profiling in the Lower Atmosphere from Wind-Induced Perturbations to Multirotor UAS," *Sensors*, Vol. 20, No. 5, 2020, p. 1,341. <https://doi.org/10.3390/s20051341>
- [2] Fuhrmann, M., and Horowitz, M. C., "Droning on: Explaining the Proliferation of Unmanned Aerial Vehicles," *International Organization*, Vol. 71, No. 2, 2017, pp. 397–418. <https://doi.org/10.1017/S0020818317000121>
- [3] Lie, F., and Gebre-Egziabher, D., "Synthetic Air Data System," *Journal of Aircraft*, Vol. 50, No. 4, 2013, pp. 1234–1249. <https://doi.org/10.2514/1.C032177>
- [4] Lee, J. H., Sevil, H., Dogan, A., and Hullender, D., "Estimation of Maneuvering Aircraft States and Time-Varying Wind with Turbulence," *AIAA Guidance, Navigation, and Control Conference*, AIAA Paper 2012-4532, 2012. <https://doi.org/10.2514/6.2012-4532>
- [5] Ahmed, Z., Halefom, M. H., and Woolsey, C. A., "Tutorial Review of Indirect Wind Estimation Methods Using Small Uncrewed Air Vehicles," *AIAA Journal of Aerospace Information Systems*, Vol. 21, No. 8, Aug. 2024, pp. 667–683. <https://doi.org/10.2514/1.1011345>
- [6] Mcmanus, H. L., and Hastings, D. E., "A Framework for Understanding Uncertainty and Its Mitigation and Exploitation in Complex Systems," *IEEE Engineering Management Review*, Vol. 34, No. 3, 2006, pp. 81–81. <https://doi.org/10.1109/EMR.2006.261384>
- [7] Kiureghian, A. D., and Ditlevsen, O., "Aleatory or Epistemic? Does It Matter?" *Structural Safety*, Vol. 31, No. 2, 2009, pp. 105–112. <https://doi.org/10.1016/j.strusafe.2008.06.020>
- [8] Smith, R. C., *Uncertainty Quantification: Theory, Implementation, and Applications*, Soc. for Industrial and Applied Mathematics, Philadelphia, PA, 2013, p. 1. <https://doi.org/10.1137/1.9781611973228>
- [9] Langelaan, J. W., Alley, N., and Neidhoefer, J., "Wind Field Estimation for Small Unmanned Aerial Vehicles," *Journal of Guidance, Control, and Dynamics*, Vol. 34, No. 4, 2011, pp. 1016–1030. <https://doi.org/10.2514/1.52532>
- [10] Kroese, D. P., Brereton, T., Taimre, T., and Botev, Z. I., "Why the Monte Carlo Method Is so Important Today," *WIREs Computational Statistics*, Vol. 6, No. 6, 2014, pp. 386–392. <https://doi.org/10.1002/wics.1314>
- [11] Danowsky, B. P., Chrstos, J. R., Klyde, D. H., Farhat, C., and Brenner, M., "Evaluation of Aeroelastic Uncertainty Analysis Methods," *Journal of Aircraft*, Vol. 47, No. 4, 2010, pp. 1266–1273. <https://doi.org/10.2514/1.47118>
- [12] Dutta, S., and Braun, R. D., "Statistical Entry, Descent, and Landing Performance Reconstruction of the Mars Science Laboratory," *Journal of Spacecraft and Rockets*, Vol. 51, No. 4, 2014, pp. 1048–1061. <https://doi.org/10.2514/1.A32937>
- [13] Albert, D. R., "Monte Carlo Uncertainty Propagation with the NIST Uncertainty Machine," *Journal of Chemical Education*, Vol. 97, No. 5, 2020, pp. 1491–1494. <https://doi.org/10.1021/acs.jchemed.0c00096>
- [14] Xiu, D., and Karniadakis, G. E., "Modeling Uncertainty in Flow Simulations via Generalized Polynomial Chaos," *Journal of Computational Physics*, Vol. 187, No. 1, 2003, pp. 137–167. [https://doi.org/10.1016/S0021-9991\(03\)00092-5](https://doi.org/10.1016/S0021-9991(03)00092-5)
- [15] Saltelli, A., Ratto, M., and Andres, T., *Introduction to Sensitivity Analysis*, Wiley, New York, 2007, pp. 1–51, Chap. 1. <https://doi.org/10.1002/9780470725184.ch1>
- [16] Sobol', I. M., "Sensitivity Estimates for Nonlinear Mathematical Models," *Mathematical Modelling and Computational Experiment*, Vol. 1, 1993, pp. 407–414. English translation of original paper in Russian (1990).
- [17] Saltelli, A., Annoni, P., Azzini, I., Campolongo, F., Ratto, M., and Tarantola, S., "Variance Based Sensitivity Analysis of Model Output. Design and Estimator for the Total Sensitivity Index," *Computer Physics Communications*, Vol. 181, No. 2, 2010, pp. 259–270. <https://doi.org/10.1016/j.cpc.2009.09.018>
- [18] Azzini, I., Mara, T. A., and Rosati, R., "Comparison of Two Sets of Monte Carlo Estimators of Sobol' Indices," *Environmental Modelling & Software*, Vol. 144, Oct. 2021, Paper 105167. <https://doi.org/10.1016/j.envsoft.2021.105167>
- [19] Etkin, B., "General Equations of Unsteady Motion," *Dynamics of Atmospheric Flight*, Wiley, New York, 1972, pp. 121–157.
- [20] Frost, W., and Bowles, R. L., "Wind Shear Terms in the Equations of Aircraft Motion," *Journal of Aircraft*, Vol. 21, No. 11, 1984, pp. 866–872. <https://doi.org/10.2514/3.45056>
- [21] González-Rocha, J., Woolsey, C. A., Sultan, C., and De Wekker, S. F. J., "Sensing Wind from Quadrotor Motion," *Journal of Guidance, Control, and Dynamics*, Vol. 42, No. 4, 2019, pp. 836–852. <https://doi.org/10.2514/1.G003542>
- [22] Simmons, B. M., Gresham, J. L., and Woolsey, C. A., "Flight-Test System Identification Techniques and Applications for Small, Low-Cost, Fixed-Wing Aircraft," *Journal of Aircraft*, Vol. 60, No. 5, 2023, pp. 1503–1521. <https://doi.org/10.2514/1.C037260>
- [23] Brandt, J., Deters, R., Ananda, G., Dantsker, O., and Selig, M., "UIUC Propeller Database, Vols 1-4," Nov. 2022, <https://m-selig.ae.illinois.edu/props/propDB.html>
- [24] Yechout, T. R., "Aircraft Equations of Motion," *Introduction to Aircraft Flight Mechanics*, AIAA Education Series, 2nd ed., AIAA Education

- Series, AIAA, Reston, VA, 2014, pp. 163–192.
<https://doi.org/10.2514/4.102547>
- [25] Morelli, E. A., and Klein, V., *Aircraft System Identification: Theory and Practice*, 2nd ed., Sunflyte Enterprises, Williamsburg, VA, 2016, pp. 181–220.
- [26] Gahan, K. C., Hopwood, J. W., and Woolsey, C. A., “Wind Estimation Using an H_∞ Filter with Fixed-Wing Aircraft Flight Test Results,” *AIAA SCITECH 2023 Forum*, AIAA Paper 2023-2252, 2023.
<https://doi.org/10.2514/6.2023-2252>
- [27] Gahan, K. C., Hopwood, J. W., and Woolsey, C. A., “Wind Estimation Using an H_∞ Filter with Fixed-Wing Aircraft Flight Test Results,” *Journal of Guidance, Control, and Dynamics*, Vol. 48, No. 3, 2025, pp. 630–637.
<https://doi.org/10.2514/1.G007735>
- [28] de Freitas Virgilio Pereira, M., “Polynomial Chaos-Based Control Design for an Aircraft at High Angles of Attack with Uncertain Aerodynamics,” Ph.D. Dissertation, Instituto Tecnológico de Aeronáutica, São José dos Campos, Brazil, 2017.
- [29] Xu, Y., “Uncertainty Quantification, State, and Parameter Estimation in Power Systems Using Polynomial Chaos Based Methods,” Ph.D. Dissertation, Virginia Polytechnic Inst. and State Univ., Blacksburg, VA, 2020.
- [30] Walters, G., “Application of Generalized Polynomial Chaos to Flow-Induced Vibration and Sound,” Ph.D. Dissertation, The Pennsylvania State Univ., University Park, PA, 2020.
- [31] Lemons, D. S., *An Introduction to Stochastic Processes in Physics*, Springer Finance Textbooks, Johns Hopkins Univ. Press, Baltimore, MD, 2002, pp. 34–35.
- [32] Xiu, D., and Karniadakis, G. E., “The Wiener–Askey Polynomial Chaos for Stochastic Differential Equations,” *SIAM Journal On Scientific Computing*, Vol. 24, No. 2, 2002, pp. 619–644.
<https://doi.org/10.1137/S1064827501387826>
- [33] Xiu, D., *Numerical Methods for Stochastic Computations: A Spectral Method Approach*, Princeton Univ. Press, Princeton, NJ, 2010, pp. 60–62.
<https://doi.org/10.5555/1893088>
- [34] McClarren, R., *Uncertainty Quantification and Predictive Computational Science: A Foundation for Physical Scientists and Engineers*, Springer International Publ., Cham, Switzerland, 2018, pp. 198–219.
- [35] Gahan, K. C., “Robust State Estimation, Uncertainty Quantification, and Uncertainty Reduction with Applications to Wind Estimation,” Ph.D. Dissertation, Virginia Polytechnic Inst. and State Univ., Blacksburg, VA, 2024.
- [36] Cox, D., Little, J., and O’Shea, D., *Ideals, Varieties, and Algorithms: An Introduction to Computational Algebraic Geometry and Commutative Algebra*, 4th ed., Springer Int’l Publishing, Switzerland, 2015, Chap. 2.
<https://doi.org/10.1007/978-3-319-16721-3>
- [37] Xiu, D., “Generalized Polynomial Chaos,” *Numerical Methods for Stochastic Computations: A Spectral Method Approach*, Preton Univ. Press, Princeton, NJ, 2010, pp. 57–67.
<https://doi.org/10.5555/1893088>
- [38] Cameron, R. H., and Martin, W. T., “The Orthogonal Development of Non-Linear Functionals in Series of Fourier-Hermite Functionals,” *Annals of Mathematics*, Vol. 48, No. 2, 1947, p. 385.
<https://doi.org/10.2307/1969178>
- [39] Debusschere, B., “Intrusive Polynomial Chaos Methods for Forward Uncertainty Propagation,” *Handbook of Uncertainty Quantification*, edited by R. Ghanem, D. Higdon, and H. Owhadi, Springer Int’l Publishing, Switzerland, 2017, pp. 617–636, Chap. 17.
<https://doi.org/10.1007/978-3-319-11259-6>
- [40] Wan, Y., Shen, D. E., Lucia, S., Findeisen, R., and Braatz, R. D., “Robust Static H_∞ Output-Feedback Control Using Polynomial Chaos,” *2018 Annual American Control Conference (ACC)*, IEEE Paper 2378-5861, 2018.
<https://doi.org/10.23919/acc.2018.8431074>
- [41] Meucci, A., *Risk and Asset Allocation*, Springer Finance Textbooks, Springer, Berlin, 2009, Chap. 2.
<https://doi.org/10.1007/978-3-540-27904-4>
- [42] Sudret, B., “Global Sensitivity Analysis Using Polynomial Chaos Expansion,” *Reliability Engineering & System Safety*, Vol. 93, No. 7, 2008, pp. 964–979.
<https://doi.org/10.1016/j.res.2007.04.002>
- [43] Alexanderian, A., Gremaud, P. A., and Smith, R. C., “Variance-Based Sensitivity Analysis for Time-Dependent Processes,” *Reliability Engineering & System Safety*, Vol. 196, April 2020, Paper 106722.
<https://doi.org/10.1016/j.res.2019.106722>
- [44] Randall, E. B., Randolph, N. Z., Alexanderian, A., and Olufsen, M. S., “Global Sensitivity Analysis Informed Model Reduction and Selection Applied to a Valsalva Maneuver Model,” *Journal of Theoretical Biology*, Vol. 526, Oct. 2021, Paper 110759.
<https://doi.org/10.1016/j.jtbi.2021.110759>
- [45] Gahan, K. C., Hopwood, J. W., and Woolsey, C. A., “Uncertainty in Wind Estimates, Part 1: Analysis Using Generalized Polynomial Chaos,” *AIAA SCITECH 2024 Forum*, AIAA Paper 2024-2824, 2024, pp. 1–15.
<https://doi.org/10.2514/6.2024-2824>
- [46] Everitt, B., and Skrondal, A., *The Cambridge Dictionary of Statistics*, Cambridge Univ. Press, Cambridge, MA, 2010, p. 89.
https://doi.org/10.1111/j.1751-5823.2011.00149_2.x
- [47] Riseborough, P., “Using the ECL EKF,” <https://docs.px4.io/main/en/advanced/config.html> [retrieved Nov. 2022].
- [48] Caflisch, R. E., “Monte Carlo and Quasi-Monte Carlo Methods,” *Acta Numerica*, Vol. 7, Jan. 1998, pp. 1–49.
<https://doi.org/10.1017/S0962492900002804>
- [49] Gahan, K. C., Hopwood, J. W., and Woolsey, C. A., “Uncertainty in Wind Estimates, Part 2: H_∞ Filtering Using Generalized Polynomial Chaos,” *AIAA SCITECH 2024 Forum*, AIAA Paper 2024-2825, 2024.
<https://doi.org/10.2514/6.2024-2825>
- [50] Navarro, M., Witteveen, J., and Blom, J., “Polynomial Chaos Expansion for General Multivariate Distributions with Correlated Variables,” arXiv, 2014.
<https://doi.org/10.48550/arXiv.1406.5483>
- [51] Rahman, S., “A Polynomial Chaos Expansion in Dependent Random Variables,” *Journal of Mathematical Analysis and Applications*, Vol. 464, No. 1, 2018, pp. 749–775.
<https://doi.org/10.1016/j.jmaa.2018.04.032>

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 Associate Editor